1 Weather

1.1 Temperature

AR process with a seasonal component.

Assume that temperature $T(t)$ at time t follows CARMA(p,q) dynamics for $q < p$, such that:

$$T(t) = \Lambda(t) + Y(t), \tag{1}$$

where $\Lambda(t)$ is a seasonality function (mean reversion function) and $Y(t) = b'(t)$.

- matrix A satisfies the conditions for stationarity
- volatility is periodic
- deseasonalized temperatures are stationary.

Benth and Benth (2012) - AR(3) with a seasonal component:

$$\Lambda(t) = a_0 + a_1 t + a_2 \cos\left(\frac{2\pi t}{365}\right) + a_3 \sin\left(\frac{2\pi t}{365}\right). \tag{2}$$

Cabrera [2010] - truncated Fourier series defined as:

$$\Lambda(t) = a_0 + a_1 t + \sum_{l=1}^{L}\left[a_{4l} \cos\left(\frac{2\pi(t-d_l)}{365}\right) + b_{4l} \sin\left(\frac{2\pi(t-d_l)}{365}\right)\right], \tag{3}$$

Truncated Fourier Series to capture the seasonal patterns evident in the time-varying variance of weather data:

$$f(t) = a_0 + \sum_{n=1}^{N}\left(a_n \cos\left(\frac{n\pi t}{365}\right) + b_n \sin\left(\frac{n\pi t}{365}\right)\right), \tag{4}$$

N - number of harmonics used in the approximation, a_n and b_n - the cosine and sine coefficients.

Final CARMA model fitted -

$$\Lambda_t = a_0 + a_1 t + \sum_{l=1}^{L} a_{l+1} \cos\left\{\frac{2\pi(t-d_l)}{365}\right\}, \tag{5}$$

Coefficients a_0 and a_1 respectively. Speed of the seasonal process is defined as $\frac{2\pi}{365}$. he CAR(p) process, which is the standard workhorse of modeling weather parameters, is a special case of the CARMA(p,q) process when q is 0.

Wind Dynamics

Benth (2010) - Under general transformation *wind speed* $W(t)$ at time $t \geq 0$:

$$W(t) = \begin{cases} [\lambda(\Lambda(t) + Y(t)) + 1]^{1/\lambda}, & \lambda \neq 0, \\ \exp(\Lambda(t) + Y(t)), & \lambda = 0, \end{cases} \tag{6}$$

for a constant λ and $\Lambda(t)$ a seasonal function as in the previous section. Benth (2010) - take

$$\frac{1}{\lambda} = 5$$

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2 Pricing

- Non-contingent contract

$$g_t(T, T) = S(T)e^{r(T-t)}, \tag{7}$$

Under real measure P in a risk-neutral measure, yielding an a-priori random price.

- Portfolio

$$\pi_t^L(\tau_1, \tau_2) + \sum_{j=1}^{L} \gamma_j W_j(\tau_1, \tau_2), \tag{8}$$

Need weight vectors for each region while optimizing the variance-covariance matrix of the payoffs.

2.1 Hedge Effectiveness

Hedged (H), Partially hedged (HP) and unhedged (P) portfolios:

$$R_H = r_t^h(\tau_1, \tau_2) + \sum_{j=1}^{L} \gamma_j W_j(\tau_1, \tau_2), \tag{9}$$

$$R_P = r_t^h(\tau_1, \tau_2), \tag{10}$$

- cost of hedging as a percentage of the revenue / real estate invetsment $\left(\frac{\bar{R}_H - \bar{R}_P}{\bar{R}_P}\right)$
- variance reduction $\left(1 - \frac{\sigma_H^2}{\sigma_P^2}\right)$